



Databasutveckling och lagring — Omtenta Normalisering

2025-05-26

Instructions

1. **No computers, no phones, no notes**
2. **No talking with other people in the room**
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. You will be presented with a database schema that isn't normalized. There are **6 normalization problems** in the given schema.
6. For each problem you need to write shortly (2 short sentences) about:
 - what the problem is
 - in which way it breaks normalization
 - how to fix it

Example:

The column X in the table Y breaks normalization by If we ..., we can solve the issue.

7. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Grading

Problems correctly recognized and solved	Grade
0 - 2	IG
3 - 5	G
6	VG

The local university is migrating to a new system, and they assigned their intern to design the new database schema. Now they've noticed there seems to be some problems with normalization, and they asked us to help them solve them.

As it turns out, the problems aren't due to the intern's lack of knowledge, the intern was in fact much more knowledgeable than it first seemed. A goodbye note was found at their empty desk, where the intern expresses their deep hate for the University, as well as admits to leaving **6 normalization errors** in the design as a *middle-finger* to the institution they so much despise.

It's up to us - as proud alumni of the University - to come to their rescue by normalizing the database.

2 Tables & sample rows

2.1 Students

StudentID	FirstName	LastName	Phones
1	Alice	Andersson	0701234567,0727654321
2	Bob	Berg	0731112222

2.2 Enrollments

StudentID	CourseID	CourseName	Credits	EnrollmentDate	Grade
1	CS101	Intro to Programming	5	2025-01-15	A
1	DB201	Databases	6	2025-01-15	B
2	CS101	Intro to Programming	5	2025-01-15	C

2.3 InstructorCourses

InstructorID	CourseID	InstructorName	InstructorPhone
10	CS101	Dr Smith	0709876543
11	DB201	Dr Karlsson	0708765432

2.4 Courses

CourseID	CourseName	DeptID	DeptName	DeptHead
CS101	Intro to Programming	D1	Computer Science	Prof Adams
DB201	Databases	D1	Computer Science	Prof Adams
MAT100	Calculus I	D2	Mathematics	Prof Brown

2.5 Rooms

RoomID	RoomNumber	BuildingID	BuildingName	BuildingAddress	Capacity
R101	101	B1	Main Hall	Campus Road 1	40
R202	202	B2	Science Center	Science Way 5	35

2.6 Books

ISBN	Title	PublisherName	PublisherAddress
978-1-11111-111-1	Database Systems	TechBooks	123 Tech Ave
978-1-22222-222-2	Intro to Programming	EduPress	45 Learning St

3 SQL code

```
CREATE TABLE Students (  
  StudentID INT PRIMARY KEY,  
  FirstName VARCHAR(50),  
  LastName VARCHAR(50),  
  Phones VARCHAR(255)  
);
```

```
CREATE TABLE Enrollments (  
  StudentID INT,  
  CourseID VARCHAR(10),  
  CourseName VARCHAR(100),  
  Credits TINYINT,  
  EnrollmentDate DATE,  
  Grade CHAR(1),  
  PRIMARY KEY (StudentID, CourseID)  
);
```

```
CREATE TABLE InstructorCourses (  
  InstructorID INT,  
  CourseID VARCHAR(10),  
  InstructorName VARCHAR(100),  
  InstructorPhone VARCHAR(15),  
  PRIMARY KEY (InstructorID, CourseID)  
);
```

```
CREATE TABLE Courses (  
  CourseID VARCHAR(10) PRIMARY KEY,  
  CourseName VARCHAR(100),  
  DeptID VARCHAR(10),  
  DeptName VARCHAR(100),  
  DeptHead VARCHAR(100)  
);
```

```
CREATE TABLE Rooms (  
  RoomID VARCHAR(10) PRIMARY KEY,  
  RoomNumber VARCHAR(10),  
  BuildingID VARCHAR(10),  
  BuildingName VARCHAR(100),  
  BuildingAddress VARCHAR(150),  
  Capacity INT  
);
```

```
CREATE TABLE Books (  
  ISBN VARCHAR(17) PRIMARY KEY,  
  Title VARCHAR(150),  
  PublisherName VARCHAR(100),  
  PublisherAddress VARCHAR(150)  
);
```

```
-- followed by the INSERT statements  
-- that match the rows above
```

4 Normalization problems

4.1 1st normalization problem

Problem is Phone column & we make new table with

(1) Student ID & Phonenummer column.

(2) Student ID $\rightarrow$ Firstname $\rightarrow$ Lastname

4.2 2nd normalization problem

There is composite ID ^{$\rightarrow$ Key} in second table due to which column

~~credits~~, Enrolment date ~~& grade~~ ^{is} are not fully dependant on the whole key

So we Break the table into two tables to satisfy 2nd Normalization

table 1 = Student ID, ^{Course ID} Credits, & grade. Second table Course ID, Course Name &

4.3 3rd normalization problem Enrollment date / we also will breakdown Enrollment date col.

Problem is Course ID column (composite key), a non-key columns are

not fully dependant of composite key so we break table into two

1- Instructor ID $\rightarrow$ Course ID

2- Instructor ID $\rightarrow$ Instructor name $\rightarrow$ Instructor Phone.

Break down Instructor ^{Name} Title, Name

4.4 4th normalization problem

Similarly, composite key & non-key columns ~~are~~ are not fully dependant

on full key, we Break the table into two, also Break down dept head column

(1) Course ID $\rightarrow$ Course Name ^{Title}

(2) Dep ID $\rightarrow$ Department Name $\rightarrow$ Dept head.

Title, Dept head

4.5 5th normalization problem

Non-key columns not fully dependant on composite key

We Break the table into two plus we can ^{also} Break down column &

(1) Room ID $\rightarrow$ Room number

(2) Building ID $\rightarrow$ Building name $\rightarrow$ Building address $\rightarrow$ capacity.

Building address
Because
two
data types
in one column.

4.6 6th normalization problem

Problem is ~~is~~ no ID column. so we ^{auto} generate ID column

we also Break down Publisher Address column because it contains Street name & Street number.

(1) Student ID $\rightarrow$ Course ID

(2) Course ID $\rightarrow$ Course Name $\rightarrow$